

Convergence Concepts in Hilbert Spaces

Strong Convergence

$$\|x_n - x\| \rightarrow 0$$

Norm convergence

Weak Convergence

$$\langle x_n, y \rangle \rightarrow \langle x, y \rangle \text{ for all } y \in H$$

Weaker notion

Weak-* Convergence

$$f_n(x) \rightarrow f(x) \text{ for all } x \in H$$

For functionals

Strong convergence $\Rightarrow$ Weak convergence, but converse is NOT true